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CONTENTS

THE CONSOLIDATED MINING AND SMELTING COMPANY OF CANADA

By F. H. MASON

The author gives a historical account of the company which owns the Tadanac smelter and the Sullivan lead-zinc-silver mine and holds a dominating position in the lead and zinc industries of the world.

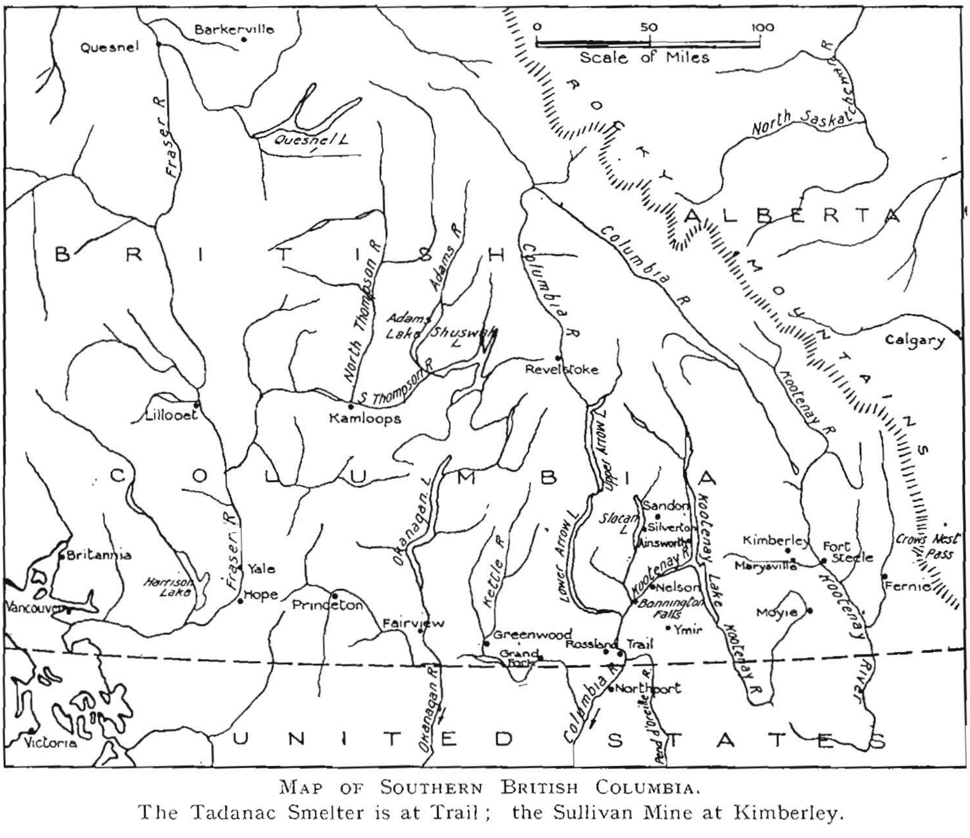
Operations at the Trail smelter, nowadays called the Tadanac smelter, were started in 1894 by F. A. Heinze, of Butte, Montana. Mr. Heinze made a contract with the Le Roi Mining Company to smelt 37,500 tons of gold-copper ore at $11 per ton, to include freight charges, and a further 37,500 tons at the lowest rate that the Le Roi Company could obtain in the open market. He obtained also a land grant from the Federal Government, in consideration for erecting a copper stack, and a bonus of $1 per ton of ore smelted. At the conclusion of the contract, the Le Roi Company, not satisfied with the rate it was receiving, decided to build a smelter of its own at Northport, Washington. The Trail concern was taken over by the British Columbia Smelting and Refining Company in 1896, which at the end of that year had five small copper furnaces in operation. Lead smelting was started in 1899. As was common with infant industries in those days, the company suffered sundry vicissitudes and underwent several reconstructions, lack of sufficient capital being one of the chief troubles. It did not get upon a really solid foundation until the Consolidated Mining and Smelting Company of Canada was formed in 1906, with an authorized capitalization of $15,000,000 to take over and operate the smelter and several mines. The Canadian Pacific Railway Company became a large shareholder, and was able to give the company moral as well as substantial backing and to help it markedly by railway expansion to open up mining areas that would bring ores to the plant.

Gold-copper ores were drawn chiefly from the company's and independent operators' mines at Rossland; copper ore from the Iron Mask mine, at Kamloops, lead ores from the company's St. Eugene mine, at Moyie, the independently owned Slocan Star, at Sandon, the Standard, at Silverton, and several smaller operators, but the real foundation of the company's prosperity was laid in 1909, when it acquired an option on the Sullivan mine, at Kimberley.

In 1896 the Sullivan group was bonded to A. Hanson, of Leadville, but the bond was not taken up. It was then bonded to Colonel Redpath and associates, who formed the Sullivan Group Mining Company. The year 1900 marked the beginning of systematic development and the first shipments of ore were made to the Hall Mines smelter at Nelson and to the Canadian smelting-works at Trail. In 1903 construction was started and a smelter (of sorts) rose into being at Marysville. The following year this smelter was entirely remodelled and, after smelting some 75,000 tons, the smelter closed down for good late in 1907. However, at the end of its smelting career, the Sullivan Group Mining Company became financially embarrassed with mining creditors, the principal one being the Crow's Nest Pass Coal Company which obtained judgment under which a seizure was made subject to the interest of the bondholders. In 1909 the bondholders and the creditors, including the Crow's Nest Pass Coal Company, reorganized the company under the name of the Fort Steele Mining and Smelting Company, the control of this company being vested in the Federal Mining and Smelting Company. In December, 1909, the Consolidated Mining and Smelting Company of Canada, Ltd., took a lease and bond on the Federal Mining and Smelting Company's holdings in the Fort Steele Mining and Smelting Company, and subsequently, towards the close of 1910, the option on the stock of the Federal Company and on that of some of the other shareholders was exercised, and the control passed into the hands of the Consolidated Mining and Smelting Company of Canada, Ltd.

The Sullivan is a curious deposit. Besides the immense body of complex intimately mixed sulphides of iron, lead, and zinc with some 6% of earthy matter, concentrations of almost clean galena, sphalerite, pyrite, and pyrrhotite occur here and there. It was these lenses of galena chiefly that were smelted in the early days, and when the Federal sold the mine, there appeared to be little prospect that the full width of the deposit would ever be treated economically. When Consolidated acquired the property, it began to explore it by diamond-drilling and soon found that it was vastly larger than the most optimistic estimate of its tonnage when the purchase was made. The present vice-president of and general manager for the company, Mr. S. G. Blaylock, informed the writer that his approximate estimate of one million tons was ridiculed at the time it was made, when the acquirement of the property was under consideration. Up to the end of 1928, 8,219,519 tons of commercial ore had been mined from the deposit. At the present time, ore is being mined at the rate of close to two million tons yearly, and estimates of the life of the mine vary at from 30 to 50 years.

Having obtained possession of the mine, the problem of treating the ore arose, for by comparison with the whole, the concentrations of shipping ore were small, and to make the mine a profitable undertaking it was realized that the whole, or nearly the whole, of the ore would have to be treated. A campaign of research followed that rarely if ever has been equalled. Commercial plants, ranging in their capacity from 50 to 600 tons of ore daily, were erected and scrapped if ineffective with an abandon that astonished the mining world. The exact cost of this work has never been made public, but it is stated that from three to five million dollars was spent on research before the present degree of efficiency was obtained. Slight differences between laboratory and commercial equipment amazingly affected results. One instance of the many may be cited. A laboratory flotation plant made of copper plate gave results approaching those desired; a commercial plant on the same lines built of wood failed. Later it was discovered that by using a small quantity of a solution of copper sulphate in the commercial operation a result equal to that of the laboratory plant was obtained. What appeared to be trifling things often made astonishing differences in results. The products from these experimental plants, even from the inefficient ones, were used to make metals. When the writer visited the smelter in 1920,



MAP OF SOUTHERN BRITISH COLUMBIA.
The Tadanac Smelter is at Trail; the Sullivan Mine at Kimberley.

one plant of 600 tons daily capacity and one of 400 tons were in operation on Sullivan ore. The larger plant was using tables and a wet magnetic separator, the latter to remove the pyrrhotite, the smaller was using tables and a modified Minerals Separation flotation plant. The products from both plants were going to the lead smelter and the electrolytic zinc plant. Both of these plants were scrapped later, but from information gained by the smaller one, the present 6,000 ton Sullivan mill was built.

In 1912 the company began its researches on the electrolytic precipitation of zinc from its solutions, and in 1915 first produced zinc commercially. Since then zinc has been added to the company's products, and the output has been steadily increased year by year. The zinc plant was in operation fully seven years before a satisfactory process for the separation of the lead and zinc mineral from the pyrrhotite gangue and from each other had been discovered, and for a time it was operated on selected ore from the mine, having a zinc content of from 15 to 18% and a lead content of 10 to 12%. This ore was crushed, roasted, the zinc extracted or partly extracted from the calcine, and the residue was sent to the lead smelter for the recovery of lead and silver.

During the War, the smelter contributed a considerable tonnage of zinc at 15·5 cents per pound for the service of the Allies when the market price for the metal in the neighbouring republic was nearly double that figure, and even at this rate the company prospered ; but after the close of the War, when base metal prices fell while the world's huge surplus stock of metals was gradually being absorbed in industry, the company fell upon hard times. At the close of the War it made the common mistake that the depression in base metal prices would be of only a short duration, believing that metals would be required to repair the ravages of War—as, of course, they were, only there was no money to pay for them—so it maintained the payment of dividends, paying them out of reserve instead of out of earnings. The error was soon realized, but the damage had been done, and a bond issue of $6,000,000 had to be floated to enable the company to carry on operations and continue its programme of expansion and research. There was no difficulty in floating the bond issue, as the potentialities of the Sullivan mine were known, and an effective, though not entirely satisfactory, process for treating the ore had been devised, so friends of the company readily subscribed for the issue. Nor did they suffer, for the majority of the $100 bonds later were retired in exchange for two shares of common stock, each of which to-day is quoted at about $400 per share and finding few sellers at that. The quotation, of course, is based on the potentialities of the shares, as during the last two years the company has paid in dividend and bonus 50% on its $25 par shares and earned between 75 and 90% on them, which would be a poor rate of interest on the quoted price. A large reserve and a contingent fund have been built up ; a good deal of which may be needed, as will be shown later.

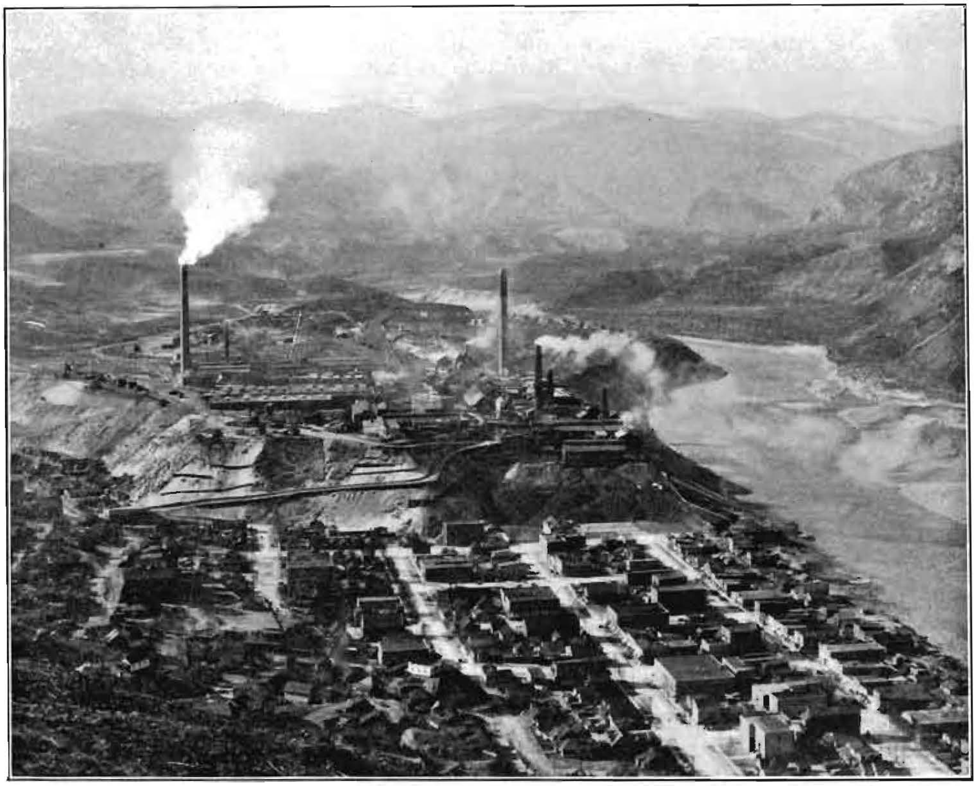
During 1920, the company brought down on itself much criticism from independent mine operators because instead of paying for ores and concentrates received it issued only warehouse receipts for the metal contents, agreeing to pay when the metal was sold. Few operators could carry on under these conditions, so the majority of the independently operated mines in the Slocan and Ainsworth districts were closed. A few concerns were sufficiently financed to allow of the continuation of development, but very few attempted to ship ore. The company's course, galling as it may have been to those affected, undoubtedly was a right one. Self preservation is a first law with companies as with individuals and if the company could not sell its own metal production it hardly could be expected to buy other people's. This condition lasted only one year.

In 1921, the company put its first zinc schedule into effect. It did not allow the company's customers much for the zinc content of their ores, contending that they should help to pay for the research that had made the treatment of zinc ore possible, but it brought about a revival of mining in the Kootenays, as most of the zinc ores of the district contain silver and the schedule provided for the payment of 80% of the silver content. The schedule has been twice amended in favour of customers ; the lead schedule also has been amended in their favour, so that, though made to share in the cost of the research, they were given the benefit of improvement in metallurgical practice as it was effected.

In June, 1922, work on the construction

of the Sullivan concentrator was started, and the plant was put into operation at the end of August, 1923. It originally was intended that the plant should have a daily capacity of 1,500 tons, but during construction the immensity of the Sullivan ore deposit was demonstrated by drilling, so it was decided to increase the size of the mill to a capacity of 3,000 tons daily and the coarse crushing plant to a capacity of between 6,000 and 7,000 tons, the latter to provide for further extension. It was at this capacity that the mill went into operation in August, 1923. Owing to the irregularity of the supply of power, which was purchased from the East Kootenay Power Company, the mill could not always be operated at this capacity. The East Kootenay Company, however, was extending its power plants to meet the requirements of the Consolidated Company, but it was realized that it would be several years before this could be accomplished, so, in 1926, when lead and zinc prices were high and increased output desired, the Consolidated Company erected an auxiliary 6,000 h.p. turbo steam plant which proved so successful that a second one was erected in the following year. Since the erection of these plants there has been no further trouble from power shortage. These auxiliary plants consist of seven Babcock and Wilcox boilers, fired by pulverized coal, and three 2,000 h.p. steam turbo generators.

The steady flow of concentrates coming from the Sullivan mill necessitated increases in the capacities of the lead and zinc plants at the smelter and of the power plant at Bonnington Falls. The last is operated by a subsidiary company, the West Kootenay Power and Light Company. In 1923 the capacity of the lead department was increased from 150 up to 225 tons of refined lead daily. This was further increased to 275 tons in 1924. The smelting plant had a greater capacity than the refining plant, so in 1924, 35,909,000 pounds of lead bullion was refined elsewhere. The development of the power plants was a slower process and, owing to a shortage of power, as the



VIEW OF TADANAC SMELTER, WITH THE TOWN OF TRAIL IN THE FOREGROUND.

zinc plant is the heaviest consumer of electric energy, the capacity of that plant was not increased proportionally. In 1924, 37,312,000 pounds of zinc was sold in the form of concentrate. Extension to the zinc plant was started in 1924 and completed in June, 1925, to synchronize with the completion of the first 20,000 h.p. unit of the new 60,000 h.p. hydro-electric station at Bonnington Falls. A second 20,000 h.p. unit was put into operation in August of the same year and a third at the end of that year.

In 1926 extensions at the Sullivan mill increased its capacity to 4,000 tons daily and the capacity of the lead and zinc plants at the smelter were still further taxed by the completion of a 500-ton mill to treat the tailing of the old gravity mill at the St. Eugene mine. These tailings had been stored in Moyie Lake, adjoining the mine, and were raised by a suction dredge for treatment. The old mill treated rather more than 1,000,000 tons of ore between 1892 and 1916—when it was burned, supposedly by enemy activity—from which about 5,800,000 ounces of silver, 225,000,000 pounds of lead, and 9,400,000 pounds of zinc were recovered. No attempt was made to save zinc until 1909. In the two and a quarter years to the end of 1928, the new flotation mill treated approximately 500,000 tons of tailing from which about 2,000 ounces of gold, 354,650 ounces of silver, 12,690,000 pounds of lead, and 15,550,000 pounds of zinc were recovered. A further addition was made to the lead plant, bringing its capacity up to 400 tons of refined lead daily and a new 100-ton addition was started at the zinc plant, to bring its capacity up to 380 tons daily. But the power question still was a difficulty and until more power was available the addition to the zinc plant could not be operated.

Another 60,000 h.p. hydro-electric plant was started on the Kootenay River at South Slocan, below Bonnington Falls, toward the end of 1926 and was put into operation in April, 1929. No notable additions were made in 1927 at either mine or smelter, the breathing time being used in improving plants and processes already in operation. In 1928 an electrolytic cadmium plant, capable of producing three-quarters of a ton of cadmium daily, an electrolytic gold refinery producing platinum and palladium as by-products, and a plant for producing bismuth were put into operation at the smelter, and work was started on the extension of the Sullivan mill to bring it up to a capacity of 6,000 tons daily. This increased capacity, which is now in operation, is not accompanied by a corresponding increase in lead and zinc production. The milling operations have been markedly improved since the mill first was put into operation and much re-modelling has been done as experience has been gained. It is now possible to treat profitably an appreciably lower grade of ore, and this is being done. The company's metal production was slightly less in the first half of 1929 than in that of 1928, but the profit of operation was greater. The metal prices, too, were greater, which probably played a part in allowing of the treating of lower-grade ore.

For several years the research department has been experimenting on a variety of processes for recovering zinc from lead smelter slag. The lead concentrates and ores that are smelted contain notable amounts of zinc and the residue from the leaching of the zinc calcine, which is smelted for its silver and lead contents, also contains zinc. Consequently, the slag from the lead smelter contains 15 to 20% of zinc. A plant consisting of two rectangular water-jacketed furnaces, in which the molten slag is treated by blowing powdered coal into it, was put into operation in August, 1929. This separates the zinc, which passes out of the furnace as fume and is cooled and collected in bag-houses. The process is said to recover 90% of the zinc content and the two furnaces have a capacity of around 60 tons of zinc daily. Some 500,000 tons of old slag, running about 20% zinc, has been accumulated.

Nothing has been said of the copper-smelting department, which initiated this huge business at Tadanac, because copper-smelting there has been gradually waning, and, though temporarily revived by the Granby Mining, Smelting, and Power Company shipping blister copper from its Anyox smelter, is likely to cease altogether when the Consolidated Company builds a copper smelter and refinery at some point yet to be decided upon on the British Columbia coast, as it has expressed its intention of doing. Should a new and important discovery of copper ore be found in south-east British Columbia the copper department would probably be retained, but at present there are no indications of such a find being made. Compared to lead and zinc the copper production of the smelter during recent years has been insignificant.

In an article of this length it would be impossible to attempt to describe the several processes used for the reduction of metals at Tadanac; a brief outline of the plant must suffice. The lead department consists of six 264 in. by 42 in. and eight 600 in. by 42 in. Dwight-Lloyd sintering machines, four 50 in. by 180 in. blast-furnaces, an anode casting wheel of 25 ton capacity per hour, two 50 ton drossing furnaces, three 60 ton kettles, one dross re-treating furnace, a 1,116 cell Betts electrolytic refinery, using 16·96 amperes per square foot. The refinery has a capacity of 400 tons of refined lead daily.

The zinc department consists of 23 Wedge mechanical roasters, each 25 ft. diameter with 9 hearths; double leaching (acid and neutral) plant, using Pachucas, Dorr tanks, Genter thickeners, mechanical agitators, American and Kelly filters, centrifugal pumps, etc.; 2,340 zinc precipitating cells, each having 17 lead anodes and 16 aluminium cathodes, using 27 to 30 amperes per square foot. Three 100 ton reverberatory furnaces melt the cathode zinc, which is cast into bars for the market. Dross from the furnaces is heat treated, leached by acid, and electrolysed. The plant has a capacity of 280 tons per day and an additional 100 ton plant will be ready to go into operation as soon as sufficient electric current is available.

Hydro-Electric Station at Lower Bonnington Falls.

The copper plant consists of a 60 ft. by 18 ft. (inside measure) reverberatory, fired by powdered coal; a 426 cell electrolytic refinery, using 17·3 amperes per square foot. The plant has a capacity of 60 tons of refined copper per day.

The silver refinery, which treats residues from the lead and copper refineries, consists of a scorifying furnace, in which impurities are removed by fuming and scorifying. The resulting doré metal is parted by sulphuric acid. The silver sulphate is diluted and the silver is precipitated as cement silver on copper, the gold is separated electrolytically from the platinum group of metals, the platinum metals are separated, and the copper sulphate resulting from the precipitation of the silver is evaporated and the sulphate crystallized out. The plant has a capacity of 20,000 ounces of silver per day and 2,500 ounces of gold per month.

The company is now installing three Brown-Boveri mercury arc transformers for converting alternating to direct current for the several electrolytic processes; up to now the conversion has been made by synchronous convertors.

Besides the metallurgical plants the smelter is equipped with a foundry, which has turned out castings up to 13½ tons, boiler shop, machine shop, blacksmith's shop, and a welding shop in which both oxy-acetylene and electric welding plants are used. This department furnished most of the plant and makes most of the repairs at the company's smelter, mines, and concentrators.

Recently the company has announced its intention of spending between seven and eight million dollars on the first unit of a synthetic fertilizer plant that will produce superphosphate of lime, ammonium sulphate, and mono- and di-ammonium phosphate, and as soon as the first unit is completed, which is scheduled for two years hence, work on a second unit will be started and will be completed within five years. It is no secret that this plant is being constructed at no desire on the part of the company, but from the necessity of finding some use for the sulphuric acid which has to be manufactured in order to remove sulphur dioxide from the smelter fume. The Canadian farmer has not learned to use fertilizers. Records indicate that Canadian agriculturists spent only slightly more than $2,000,000 on fertilizers in 1928, or about two-fifths of what Canada exported. The record is not encouraging to a company starting on a large fertilizer-manufacturing enterprise. But claims against the company for damage from smelter fume, particularly in the neighbouring State, Washington, had been piling up; an international commission was convened to investigate the claims and it was to this commission that the Consolidated Company presented its plans for at any rate taking the sting out of the fume. Each unit of the projected plant will have a contact sulphuric acid plant capable of making 300 tons of acid daily, which gives some idea of the amount of acid fume that is going into the atmosphere.

The construction of this plant together with the settlements of several hundred claims for damages done by smelter fume is likely to make a considerable inroad on the contingency fund that the company has so wisely built up out of excess profits during recent years.

Since the smelter was started, in 1894, to the end of 1928, it has treated 14,256,036 tons of material and has produced 2,188,039 ounces of gold, 69,364,136 ounces of silver, 2,088,030,000 pounds of lead, 873,422,877 pounds of zinc, and 160,541,324 pounds of copper. The output in 1928 was 1,667,586 tons of material, 23,623 ounces of gold, 7,673,762 ounces of silver, 318,831,578 pounds of lead, 163,530,890 pounds of zinc, and 17,806,306 pounds of copper.

The electric energy for the company's operations is provided by three plants on the Kootenay River, two at Bonnington Falls and one at South Slocan, situated between 13 and 15 miles below Nelson and operated by a subsidiary company, the West Kootenay Power and Light Company, which, besides providing current for the smelter supplies the Granby Consolidated Mining, Smelting and Power Company's Copper Mountain mine and Allenby mill and many towns and settlements en route. It also intends to erect a transmission line into the Slocan district. The stations have rated capacities of 34,000, 60,000 and 60,000 h.p., respectively, capacities that vary considerably with the season, a condition that the company hopes to remove by the erection of a dam across the exit of Kootenay Lake to raise the low-water level six feet. Permission is now being sought for the erection of this dam. The company has received permission from the Provincial Government for the construction of an 80,000 h.p. hydro-electric station on the Pend d'Oreille River, near the international boundary, and a 30,000 h.p. plant on the Adams River, 35 miles north-east of Kamloops. It is understood that the whole of the power from these stations has already been allocated. As soon as the preliminary work has been completed the construction of these new plants will be pushed energetically, but, of course, before the construction of these big power plants a good deal of preliminary work is necessary.

Turning from the metallurgical to the mining operations of the company, in the early days gold-copper ore was obtained chiefly from the company's group of mines at Rossland, of which the War Eagle, Centre Star, and Le Roi were the chief, and from independent operators in the same camp, and silver-lead ore from the St. Eugene, at Moyie, and from independent operators in the Slocan and Ainsworth districts. In 1910 silver-lead ore began to come from the Sullivan, and by the time the St. Eugene had become exhausted, in 1915, the Sullivan was sending from 45,000 to 50,000 tons per annum to the smelter. The Rossland mines began to show signs of exhaustion before the end of the war; there still was a large ore reserve in them, but development failed to bring in anything like as much new ore as was mined each year. In 1916 Consolidated acquired a controlling interest in the Coast Copper Company which owns the Old Sport mine in the northern part of Vancouver Island and in 1919 control of Sunloch mines, which had developed a considerable tonnage of copper ore at Jordan River, in the southern end of Vancouver Island. After the war ceased and Government-fixed prices were removed, the price of the metal soon dropped to figures that made these properties uninteresting for the time being and their development was not pushed. Moreover, the company needed all its resources for the exploitation of the Sullivan. The company's power subsidiary, however, extended its power line to the Canada Copper Corporation's mine at Copper Mountain and mill at Allenby and the Canadian Pacific Railway built the branch line from Princetown to Copper Mountain by way of Allenby, thus providing feed for the copper department at the smelter.

But the best laid plans miscarry at times. The world was in a state of unrest after the war and labour in particular was fractious. There were strikes in the construction of the branch line, strikes in the operation of the mine, and strikes in the construction of the mill, so that by the time Canada Copper was ready to ship the price of copper had fallen to 13 cents. A going concern might have operated at this price, but a new concern hampered by a heavy bonded indebtedness could not, so after shipping concentrate to Tadanac for a few months Canada Copper closed the mine. It had to pass interest on its bonds, and the bondholders took possession. Allenby Copper Corporation was formed as a go-between to take over the properties and transfer them to the Granby Consolidated Mining, Smelting and Power Company. Granby remodelled the mill and shipped to Tadanac until the end of August, 1928, when the contract entered into between Canada Copper and Consolidated, which Granby took over, expired. Granby sought better smelting and power rates, and, failing to get them, diverted concentrate to the Tacoma smelter.

Seeing how the wind was blowing, and well provided with funds from the profits of its Sullivan mine, Consolidated started a vigorous search for a copper property of its own, and if in that search other properties were found and could be secured on reasonable terms they were not neglected because they happened not to be copper deposits. The result is that to-day the company owns or controls or has options on properties not only throughout British Columbia, but in every Province in Canada with the exception of Prince Edward Island, in Newfoundland, Yukon Territory, and the United States. Options are taken up so often and dropped equally often if proved wanting that it is impossible to keep track of the company's activities in British Columbia and quite hopeless to attempt to follow them beyond; and, for sufficiently good reasons, the company tells the public only what it considers is good for the public to know. Nor does it give information as to the ore reserves at its several properties, costs of operations, and recoveries. The only variation from this rule that I know of is that the cost of mining at the Sullivan is approximately 25 cents per ton, an achievement evidently considered sufficiently remarkable to justify the exception.

The company's major mining operation in British Columbia, of course, is at the Sullivan, a description of which was given in a paper, read recently before the Western Branch of the Canadian Institute of Mining and Metallurgy, by Mr. D. L. Thompson, assistant superintendent. The next largest operation is at the Coast Copper Company's Old Sport, where undoubtedly a considerable tonnage of ore running around 3% copper has been developed above the 1,250 ft. level. The company is now sinking below that level to develop at greater depth. A good tonnage of copper ore has been developed at the Sunloch mine, but little work has been done there during recent years. A vigorous diamond-drilling campaign is being carried on at the George group, in the Bear River section of the

Portland Canal division, and some minable bodies of copper and copper-zinc ore have been penetrated. An effort is now under way to open these by a tunnel. A large amount of exploration has been and is being done at the Big Missouri group. Here rich unrelated narrow lenses of gold ore have been found and an immense body of low-grade material, so low grade that as yet it is not certain that it is ore. Much more work will have to be done before the value—if it has value—of this property can be determined. The Rock Candy fluorspar mine has been and may again be a mine of considerable importance. It produces a high-grade fluorite concentrate, large quantities of which were shipped to Gary, Indiana, prior to the Fordeney tariff going into effect. The mine is now operated intermittently to provide fluorspar for the manufacture of hydrofluosilicic acid which are used as an electrolyte in refining lead bullion by the Betts method at the smelter. The phosphate of lime properties are rapidly becoming important operations and will become more and more so as the fertilizer plant is developed. They are located in that immense belt, croppings of which have been found here and there from Banff south through Idaho and Montana into the States of Utah and Colorado. It is from this belt in Idaho that the Anaconda Copper Mining Company has been quarrying phosphate rock for the manufacture of superphosphate for several years. The quality appears to be appreciably higher in Idaho than in British Columbia, but much more development has been done there. The only other major operation in British Columbia is at the Hunter V mine, at Ymir, which in 1928 produced 10,500 tons of low-grade siliceous silver ore which is used for fluxing at the smelter. The other properties in British Columbia are either old mines, which for the time being are either closed or have been turned over to lessees, or prospects. The St. Eugene mill is still operating at capacity—500 tons daily—on the old tailing, but it is expected that it will clean up the accumulation by the end of the year. It will then be used to treat oxidized ore from the Sullivan dumps, thereby not allowing this material to interfere with the smooth operation of the Sullivan mill, the oxidized ore requiring a different treatment from the freshly mined ore.

The company's prospecting activities in the Province are widely distributed from the Kootenays to the Portland Canal. Most of the exploration is being done by diamond-drilling but surface and underground work are being done at several of the properties in the Omineca division, where several properties give promise of being developed into producers.

In the spring of 1929 the company acquired by purchase the Canadian North-eastern Railway from Vancouver holdings. The line at present has been completed for 14 miles up the Bear River Valley from the town of Stewart, but is now in bad condition. The purchase included a charter giving the right to extend the line to the Finlay River, and with a branch from the main line to the Yukon boundary. Two survey parties have been in the field since the late spring locating a route for the extension. The company has made no announcement as to the purpose to which it proposes to put the railway. It must be remembered, however, that the control of the company's stock is held by the Canadian Pacific Railway and its friends, and it is not unlikely the Consolidated is acting only as an agent for the larger concern in the matter. The railway situation in Canada is complicated, and it is possible the Consolidated company might more easily and cheaply obtain a concession than the Canadian Pacific Railway.

What of the men who built up this organization ? As the present vice-president and general manager so often has emphazized, it has been the result of team work, and and as the team is a large one it is impossible to mention each individual. But a team is not likely to achieve good results without an able captain, and this team seems to have been blessed with unusually capable captains. Messrs. W. H. Aldridge and R. H. Stewart did some splendid work on the foundations. The present skipper, Mr. S. G. Blaylock, stands out pre-eminently, as he guided the company through the troubled waters of post-war days and brought it to its present enviable position. Mention should be made, too, of Mr. W. M. Archibald, chief mining engineer, for his splendid work at the Sullivan mine, and to Mr. J. J. Warren, president, for the skilful way in which he has handled the finances of the company.

I desire to thank Mr. J. D. Galloway, Provincial Mineralogist for British Columbia, for reading this article and for the suggestions he made for its improvement.